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Studierichting: Informatica
Oefeningenreeks 2B nr. 6.5

$\alpha = \text{Bgtan}\,\frac{1}{7}$

$\beta = \text{Bgtan}\,\frac{1}{3}$

$\alpha + 2\beta$

$\tan\alpha = \frac{1}{7}$

Dubbele hoekformule: $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

$$\tan 2\beta = \frac{2\tan\beta}{1-\tan^2\beta} = \frac{2\cdot\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2}$$
$$= \frac{\frac{2}{3}}{1-\frac{1}{9}} = \frac{\frac{2}{3}}{\frac{9-1}{9}} = \frac{\frac{2}{3}}{\frac{8}{9}}$$
$$= \frac{2}{3}\cdot\frac{9}{8} = \frac{18}{24} = \frac{3}{4}$$

$\Rightarrow \tan(\alpha + 2\beta)$

Somformule: $\tan(A+B) = \frac{\tan A + \tan B}{1-\tan A\cdot\tan B}$

$$= \frac{\tan\alpha + \tan 2\beta}{1-\tan\alpha\cdot\tan 2\beta}$$
$$= \frac{\frac{1}{7}+\frac{3}{4}}{1-\frac{1}{7}\cdot\frac{3}{4}}$$

Teller: $\frac{1}{7}+\frac{3}{4} = \frac{4}{28}+\frac{21}{28} = \frac{25}{28}$

Noemer: $1-\frac{1}{7}\cdot\frac{3}{4} = 1-\frac{3}{28} = \frac{28-3}{28} = \frac{25}{28}$

$$= \frac{\frac{25}{28}}{\frac{25}{28}} = 1$$

References